

Inventory Record #112

Simultaneous truth and performance level estimation (STAPLE): an algorithm for the validation of image segmentation

Authors

Warfield SK, Zou KH, Wells WM.

Published in

IEEE Transactions on Medical Imaging, vol. 23, no. 7, pp. 903-921, 2004

DOI

10.1109/TMI.2004.828354

Source

<https://ieeexplore.ieee.org/document/1309714/>

Why the full text is not held

IEEE Xplore subscription required. An author manuscript is available at PMC3788853 (PMID 15250643) if institutional access is unavailable.

Role in this programme

The standard method for fusing several raters into one reference standard. Relevant to Chapter 3 wherever multiple readers contribute to the reference standard, and to the harmonised central-canal regrading proposed for external validation.

ABSTRACT

Characterizing the performance of image segmentation approaches has been a persistent challenge. Performance analysis is important since segmentation algorithms often have limited accuracy and precision. Interactive drawing of the desired segmentation by human raters has often been the only acceptable approach, and yet suffers from intra-rater and inter-rater variability. Automated algorithms have been sought in order to remove the variability introduced by manual raters, but such algorithms must be assessed to ensure they are suitable for the task. The performance of raters (human or algorithmic) generating segmentations of medical images has been difficult to quantify because of the difficulty of obtaining or estimating a known true segmentation for clinical data. The authors present an expectation-maximization algorithm for simultaneous truth and performance level estimation (STAPLE). The algorithm considers a collection of segmentations and computes a probabilistic estimate of the true segmentation and a measure of the performance level represented by each segmentation.

Abstract reproduced from the publisher record for identification and cataloguing. Generated for the Rizgary lumbar spine research programme (Dr. Polla Fattah). Obtain the full text before citing substantive content.